21.5: Duplication, rearrangement, and mutation of DNA contribute to genome evolution

EVOLUTION Now that we have explored the makeup of the human genome, let's see what its composition reveals about how the genome evolved. The basis of change at the genomic level is mutation, which underlies much of genome evolution. It seems likely that the earliest forms of life had a minimal number of genes—those necessary for survival and reproduction. If this were indeed the case, one aspect of evolution must have been an increase in the size of the genome, with the extra genetic material providing the raw material for gene diversification. In this section, we'll first look at how extra copies of all or part of a genome can arise and then consider subsequent processes that can lead to the evolution of proteins (or RNA products) with slightly different or entirely new functions.

Duplication of Entire Chromosome Sets

An accident in meiosis, such as failure to separate homologs during meiosis I, can result in one or more extra sets of chromosomes, a condition known as polyploidy (see Concept 15.4). Although such accidents would most often be lethal, in rare cases they could facilitate the evolution of genes. In a polyploid organism, one set of genes can provide essential functions for the organism. The genes in the one or more extra sets can diverge by accumulating mutations; these variations may persist if the organism carrying them survives and reproduces. In this way, genes with novel functions can evolve. As long as one copy of an essential gene is expressed, the divergence of another copy can lead to its encoded protein acting in a novel way, thereby changing the organism's phenotype.

The outcome of this accumulation of mutations may eventually be the branching off of a new species. While polyploidy is rare among animals, it is relatively common among plants, especially flowering plants. Some botanists estimate that as many as 80% of the plant species that are alive today show evidence of polyploidy having occurred among their ancestral species. You'll learn more about the details of how polyploidy leads to plant speciation in Concept 24.2.

Alterations of Chromosome Structure

With the recent explosion in genomic sequence information, we can now compare the chromosomal organizations of many different species in detail. This information allows us to make inferences about the

evolutionary processes that shape chromosomes and may drive speciation. For example, scientists have long known that sometime in the last 7–8 million years, when the ancestors of humans and chimpanzees diverged as species, the fusion of two ancestral chromosomes in the human line led to different haploid numbers for humans (n=23) and chimpanzees (n=24). The banding patterns in stained chromosomes suggested that the ancestral versions of current chimpanzee chromosomes 12 and 13 fused end to end, forming chromosome 2 in an ancestor of the human lineage **(Figure 21.11)**.

Figure 21.11

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Human and chimpanzee chromosomes.
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The telomere, centromere, telomere-like, and centromere-like sequences on human chromosome 2 match the equivalent regions on chimpanzee chromosomes 12 and 13.

The positions of telomere-like and centromere-like sequences on human chromosome 2 (left) match those of telomeres on chimpanzee chromosomes 12 and 13 and the centromere on chimpanzee chromosome 13 (right). This suggests that chromosomes 12 and 13 in a human ancestor fused end to end to form human chromosome 2. The centromere from ancestral chromosome 12 remained functional on human chromosome 2, while the one from ancestral chromosome 13 did not.

How do we know a chromosome didn't just split into two in an ancestor of chimps? A comparison of chromosomes in other great apes—gorillas, chimps, bonobos, and orangutans, our closest relatives—shows that those species have 24 chromosomes, so the simplest conclusion is that the last common ancestor of all great apes had 24 chromosomes. (You'll learn more about this type of reasoning in Concept 26.3.) Sequencing and analysis of human chromosome 2 during the Human Genome Project revealed sequences for telomeres and an extra, unused centromere in the middle of it, among other very strong supporting evidence for the model described previously (see Figure 21.11).

In another study of broader scope, researchers compared the DNA sequence of each human chromosome with the whole-genome sequence of the mouse **(Figure 21.12)**. One part of their study showed that large blocks of genes on human chromosome 16 are found on four mouse chromosomes, indicating that the genes in each block stayed together in both the mouse and the human lineages during their divergent evolution from a common ancestor.

Figure 21.12

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Human and mouse chromosomes.
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Human chromosome 16 has parts that match up with regions of mouse chromosomes 7, 8, 16, and 17.

Here, we can see that DNA sequences very similar to large blocks of human chromosome 16 (colored areas in this diagram) are found on mouse chromosomes 7, 8, 16, and 17. This finding suggests that the DNA sequence in each block has stayed together in the mouse and human lineages since the time they diverged from a common ancestor.

Interview

Interview with Eric Lander: Exploring the human genome

Eric Lander.

Performing the same comparison of chromosomes of humans and six other mammalian species allowed the researchers to reconstruct the evolutionary history of chromosomal rearrangements in these eight species. They found many duplications and inversions of large portions of chromosomes, the result of errors during meiotic recombination in which the DNA was broken and rejoined incorrectly. The rate of these events seems to have begun accelerating about 100 million years ago, around 35 million years before large dinosaurs became extinct and the number of mammalian species began rapidly increasing. The apparent coincidence is interesting because chromosomal rearrangements are thought to contribute to the generation of new species. Although two individuals with different arrangements could still mate and produce offspring, the offspring would have two nonequivalent sets of chromosomes, making meiosis inefficient or even impossible. Thus, chromosomal rearrangements would lead to two populations that could not successfully mate with each other, a step on the way to their becoming two separate species. (You'll learn more about this in Concept 24.2.)

The same study also unearthed a pattern with medical relevance. Analysis of the chromosomal breakage points associated with the rearrangements showed that specific sites were used over and over again. A number of these recombination "hot spots" correspond to locations of chromosomal rearrangements within the human genome that are associated with congenital diseases (see Concept 15.4).

Duplication and Divergence of Gene-Sized Regions of DNA

Errors during meiosis can also lead to the duplication of chromosomal regions that are smaller than the ones we've just discussed, including segments the length of individual genes. Unequal crossing over during prophase I of meiosis, for instance, can result in one chromosome with a deletion and another with a duplication of a particular gene or genes. Transposable elements can provide homologous sites where

nonsister chromatids can cross over, even when other chromatid regions are not correctly aligned **(Figure 21.13)**.

Figure 21.13

Gene duplication due to unequal crossing over.

Gene duplication due to unequal crossing over.

One mechanism by which a gene (or other DNA segment) can be duplicated is recombination during meiosis between copies of a transposable element (yellow) flanking the gene (blue). Such recombination between misaligned nonsister chromatids of homologous chromosomes produces one chromatid with two copies of the gene and one chromatid with no copy. (Genes and transposable elements are shown only in the region of interest.)

Duplicated chromosomes (sister chromatids) in homologous pair. Gene and transposable element labeled. Non-sister chromatids (dark purple chromosome above light purple chromosome) lined up asymmetrically at the crossover point (transposable element) with the dark purple chromosome above the light purple chromosome. Incorrect pairing of two homologs during meiosis leads to one chromatid with two copies of the gene and one chromatid with no copy.

MAKE CONNECTIONS *Examine how crossing over occurs in*Figure 13.9*. In the middle panel above, draw a line through the portions that result in the upper chromatid in the bottom panel. Use a different color to do the same for the other chromatid.*

*For suggested answer, see*Appendix A*.*

Also, slippage can occur during DNA replication, such that the template shifts with respect to the new complementary strand, and a part of the template strand is either skipped by the replication machinery or used twice as a template. As a result, a segment of DNA is deleted or duplicated. It is easy to imagine how such errors could occur in regions of repeats. (See the question in Figure 21.13.) The variable number of repeated units of simple sequence DNA at a given site, used for STR analysis, is probably due to errors like these. Evidence that unequal crossing over and template slippage during DNA replication lead to duplication of genes is found in the existence of multigene families, such as the globin family.

Evolution of Genes with Related Functions: The Human Globin Genes

In Figure 21.10b, you saw the organization of the α-globin and β-globin gene families as they exist in the human genome today. Now, let's consider how events such as duplications can lead to the evolution of genes with related functions like the globin genes. A comparison of gene sequences within a multigene family can suggest the order in which the genes arose. Re-creating the evolutionary history of the globin genes using

this approach indicates that they all evolved from one common ancestral globin gene that underwent duplication and divergence into the α-globin and β-globin ancestral genes about 450–500 million years ago **(Figure 21.14)**. Each of these genes was later duplicated several times, and the copies then diverged from each other in sequence, yielding the current family members. In fact, the common ancestral globin gene also gave rise to the oxygen-binding muscle protein myoglobin and to the plant protein leghemoglobin. The latter two proteins function as monomers, and their genes are included in a "globin superfamily."

Figure 21.14

A proposed model for the sequence of events in the evolution of the human α-globin and β-globin gene families from a single ancestral globin gene.

Evolutionary tree as a model for the evolution of human alpha-globin and beta-globin from a single ancestral globin gene.

Evolutionary tree as a model for the evolution of human alpha-globin and beta-globin from a single ancestral globin gene. The alpha-globin gene family is on Chromosome 16 and the beta-globin gene family is on Chromosome 11.

Key: dark blue = gene, light blue = noncoding region, gold = pseudogenes.

Genes and pseudogenes are named with Greek letters. Each step moves forward in evolutionary time. Top of the tree (furthest back in evolutionary time) = ancestral globin gene. Duplication of the ancestral gene leads to gene, noncoding region, gene. Mutation in both copies leads to the alpha gene; noncoding region, beta gene. Transposition to different chromosomes leads to two branches in the tree: One headed by alpha gene, the other headed by beta gene. Further duplications and mutations lead to the alpha and beta globin gene families on chromosomes 16 and 11, respectively. Alpha gene branch: one branch to zeta gene and one branch to alpha gene. Zeta gene: one branch leads to zeta gene and another to psi zeta gene (both on chromosome 16). Alpha gene: branches to psi alpha 2 pseudogene, psi alpha 1 pseudogene, alpha 2 gene, alpha 1 gene, and gamma theta pseudogene (all on chromosome 16). Beta gene branch: branches to epsilon gene, gamma gene, beta gene. Epsilon gene: goes to epsilon gene on chromosome 11. Gamma gene: branches to gamma G gene and gamma A gene on chromosome 11. Beta gene branches to psi beta pseudogene, delta gene and beta gene, all on chromosome 11.

The gold elements are pseudogenes. Explain how they could have arisen after gene duplication.

*For suggested answer, see*Appendix A*.*

After the duplication events, the differences between the genes in the globin families undoubtedly arose from mutations that accumulated in the gene copies over many generations. The current model is that the necessary function provided by an α-globin protein, for example, was fulfilled by one gene, while other

copies of the α-globin gene accumulated random mutations. Many mutations may have had an adverse effect on the organism, and others may have had no effect. However, a few mutations must have altered the function of the protein product in a way that benefitted the organism at a particular life stage without substantially changing the protein's oxygen-carrying function. Presumably, natural selection acted on these altered genes, maintaining them in the population.

In the **Scientific Skills Exercise**, you can compare amino acid sequences of the globin family proteins and see how such comparisons were used to generate the model for globin gene evolution shown in Figure 21.14. The existence of several pseudogenes among the functional globin genes provides additional evidence for this model: Random mutations in these "genes" over evolutionary time have destroyed their function.

Scientific Skills Exercise

Reading an Amino Acid Sequence Identity Table

AP Science Practices 2, 4, 5

How Have Amino Acid Sequences of Human Globin Genes Diverged During Their Evolution? To build a model of the evolutionary history of the globin genes (see Figure 21.14), researchers compared the amino acid sequences of the polypeptides they encode. In this exercise, you will analyze comparisons of the amino acid sequences of the globin polypeptides to shed light on their evolutionary relationships.

A 3 D model of hemoglobin.

Hemoglobin

How the Experiment Was Done Scientists obtained the DNA sequences for each of the eight globin genes and "translated" them into amino acid sequences. They then used a computer program to align the sequences (with dashes indicating gaps in one sequence) and calculate a percent identity value for each pair of globins. The percent identity reflects the number of positions with identical amino acids relative to the total number of amino acids in a globin polypeptide. The data were displayed in a table to show the pairwise comparisons.

Data from the Experiment The following table shows an example of a pairwise alignment—that of the α1-globin (alpha-1globin) and ζ-globin (zeta globin) amino acid sequences—using the standard single-letter symbols for amino acids. To the left of each line of amino acid sequence is the number of the first amino acid in that line. The percent identity value for the α1- and ζ-globin amino acid sequences was calculated by counting the number of matching amino acids (86, highlighted in yellow), dividing by the total number of

amino acid positions (143), and then multiplying by 100. This resulted in a 60% identity value for the α1-ζ pair, as shown in the amino acid identity table below. The values for other globin pairs were calculated in the same way.

A chart matches the single-letter amino acids in an alignment of globin amino acid sequences.

Globin	Alignment of globin amino acid sequences
alpha 1	1 M V L S P A D K T N V K A A W G K V G A H A G E Y G A E A L
zeta	1 M S L T K T E R T I I V S M W A K I S T Q A D T I G T E T L
alpha 1	31 E R M F L S F P T T K T Y F P H F D L S H - G S A Q V K G H
zeta	31 E R L F L S H P Q T K T Y F P H F D L - H P G S A Q L R A H
alpha 1	61 G K K V A D A L T N A V A H V D D M P N A L S A L S D L H A
zeta	61 G S K V V A A V G D A V K S I D D I G G A L S K L S E L H A
alpha 1	91 H K L R V D P V N F K L L S H C L L V T L A A H L P A E F T
zeta	Y I L R V D P V N F K L L S H C L L V T L A A R F P A D F T
alpha 1	121 P A V H A S L D K F L A S V S T V L T S K Y R
zeta	121 A E A H A A W D K F L S V V S S V L T E K Y R

An amino acid identity table is given with rows labeled from top to bottom, and columns labeled from left to right, as follows (for each): alpha 1, alpha 2, zeta, beta, delta, epsilon, gamma A, and gamma G.

Compiled using data from the National Center for Biotechnology Information (NCBI).

Amino acid identity table
alpha family
alpha family
alpha 2
zeta
beta family

delta
epsilon
gamma A
gamma G

Further Reading R. C. Hardison, Globin genes on the move, *Journal of Biology* 7:35.1–35.5 (2008).

INTERPRET THE DATA

- Note that in the amino acid identity table, the data are arranged so each globin pair can be compared.
- Some cells in the table have dashed lines. Given the pairs that are being compared for these cells, what percent identity value is implied by the dashed lines?
- Notice that the cells in the lower left half of the table are blank. Using the information already provided in the table, fill in the missing values. Why does it make sense that these cells were left blank in the table?
- The earlier that two genes arose from a duplicated gene, the more their nucleotide sequences can have diverged, which may result in amino acid differences in the protein products.
- Based on that premise, identify which two genes are most divergent from each other. What is the percent amino acid identity between their polypeptides?
- Using the same approach, identify which two globin genes are the most recently duplicated. What is the percent identity between them?
- The model of globin gene evolution shown in Figure 21.14 suggests that an ancestral gene duplicated and mutated to become α- and β-globin genes, and then each one was further duplicated and mutated. What features of the data set support the model?
- Make an ordered list of all the percent identity values from the table, starting with 100% at the top. Next to each number write the globin pair(s) with that percent identity value. Use one color for the globins from the α family and a different color for the globins from the β family.
- Compare the order of pairs on your list with their positions in the model shown in Figure 21.14. Does the order of pairs describe the same relative “closeness” of globin family members seen in the model?

- Compare the percent identity values for pairs within the α or β family. to the values for between-family pairs.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Evolution of Genes with Novel Functions

In the evolution of the globin gene families, gene duplication and subsequent divergence produced family members whose protein products performed similar functions (oxygen transport). Alternatively, one copy of a duplicated gene can undergo alterations that lead to a completely new function for the protein product. The genes for lysozyme and α-lactalbumin are a good example.

Lysozyme is an enzyme that helps protect animals against bacterial infection by hydrolyzing bacterial cell walls (see Visualizing Figure 5.16); α-lactalbumin is a nonenzymatic protein that plays a role in milk production in mammals. The two proteins are quite similar in their amino acid sequences and three-dimensional structures **(Figure 21.15)**. Both genes are found in mammals, but only the lysozyme gene is present in birds. These findings suggest that at some time after the lineages leading to mammals and birds had separated, the lysozyme gene was duplicated in the mammalian lineage but not in the avian lineage. Subsequently, one copy of the duplicated lysozyme gene evolved into a gene encoding α-lactalbumin, a protein with a completely new function associated with a key characteristic of mammals—milk production. In one study, evolutionary biologists searched vertebrate genomes for genes with similar sequences. There appear to be at least eight members of the lysozyme family, distributed widely among mammalian species. The functions of all the encoded gene products are not yet known, but it will be exciting to discover whether they are as different as the functions of lysozyme and α-lactalbumin.

Figure 21.15

Comparison of lysozyme and α-lactalbumin proteins.

Computer models of a lysozyme and b alpha-lactalbumin are similar ribbon and string-like shapes. Amino acid sequence alignments of the molecules show 3 sets of rows with highlighted single-letter amino acids that match.

Computer-generated ribbon models of the similar structures of **(a)** lysozyme and **(b)** α-lactalbumin are shown, along with **(c)** a comparison of the amino acid sequences of the two proteins. Single-letter amino acid codes are used (see Figure 5.14). Identical amino acids are highlighted in yellow, and dashes indicate gaps in one sequence that have been introduced by the software to optimize the alignment.

MAKE CONNECTIONS *Even though two amino acids are not identical, they may be structurally and chemically similar and therefore behave similarly. Using* ***Figure 5.14*** *as a reference, examine the*

nonidentical amino acids in positions 1–30 and note cases where the amino acids in the two sequences are similarly acidic or basic.

*For suggested answer, see*Appendix A.**

Besides the duplication and divergence of whole genes, rearrangement of existing DNA sequences within genes has also contributed to genome evolution. The presence of introns may have promoted the evolution of new proteins by facilitating the duplication or shuffling of exons, as we'll see next.

Rearrangements of Parts of Genes: Exon Duplication and Exon Shuffling

Recall from Concept 17.3 that an exon often codes for a protein domain, a distinct structural and functional region of a protein molecule, such as the WD40 domain in Figure 21.3. We've already seen that unequal crossing over during meiosis can lead to duplication of a gene on one chromosome and its loss from the homologous chromosome (see Figure 21.13). By a similar process, a particular exon within a gene could be duplicated on one chromosome and deleted from the other. The gene with the duplicated exon would code for a protein containing a second copy of the encoded domain. This change in the protein's structure might augment its function by increasing its stability, enhancing its ability to bind a particular ligand, or altering some other property. Quite a few protein-coding genes have multiple copies of related exons, which presumably arose by duplication and then diverged. The gene encoding the extracellular matrix protein collagen is a good example. Collagen is a structural protein (see Figure 5.18) with a highly repetitive amino acid sequence, which reflects the repetitive pattern of exons in the collagen gene.

As an alternative possibility, we can imagine the occasional mixing and matching of different exons either within a gene or between two different (nonallelic) genes owing to errors in meiotic recombination. This process, termed *exon shuffling*, could lead to new proteins with novel combinations of functions. As an example, let's consider the gene for tissue plasminogen activator (TPA). The TPA protein is an extracellular protein that helps control blood clotting. It has four domains of three types, each encoded by an exon, and one of those exons is present in two copies. Because each type of exon is also found in other proteins, the current version of the gene for TPA is thought to have arisen by several instances of exon shuffling during errors in meiotic recombination and subsequent duplication **(Figure 21.16)**.

Figure 21.16

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Evolution of a new gene by exon shuffling.
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Diagram of portions of ancestral genes moving into evolving gene for TPA due to exon shuffling.

Meiotic errors could have moved exons, each encoding a particular domain, from ancestral forms of the genes for epidermal growth factor, fibronectin, and plasminogen (left) into the evolving gene for tissue plasminogen activator, TPA (right). Subsequent duplication of the "kringle" exon (K) from the plasminogen gene after its movement into the TPA gene could account for the two copies of this exon in the TPA gene existing today.

Diagram of portions of ancestral genes moving into evolving gene for TPA due to exon shuffling. Portions of ancestral genes are as follows:

1. Epidermal growth factor gene with multiple EGF exons.
2. Fibronectin gene with multiple finger exons.
3. Plasminogen gene with a kringle exon.

An exon from each of the three ancestral portions is shuffled into the TPA gene as it exists today, diagrammed here as follows: Non-coding region; finger exon; non-coding region; kringle exon; non-coding region; another kringle exon due to exon duplication.

VISUAL SKILLS *Looking at*Figure 21.13*, describe the steps by which transposable elements within introns might have facilitated the exon shuffling shown here.*

*For suggested answer, see*Appendix A*.*

How Transposable Elements Contribute to Genome Evolution

The persistence of transposable elements as a large fraction of some eukaryotic genomes is consistent with the idea that they play an important role in shaping a genome over evolutionary time. These elements can contribute to the evolution of the genome in several ways. They can promote recombination, disrupt cellular genes or control elements, and carry entire genes or individual exons to new locations.

Transposable elements of similar sequence scattered throughout the genome facilitate recombination between different (nonhomologous) chromosomes by providing homologous regions for crossing over (see Figure 21.13). Most such recombination events are probably detrimental, causing chromosomal translocations and other changes in the genome that may be lethal to the organism. But over the course of evolutionary time, an occasional recombination event of this sort may be advantageous to the organism. (For the change to be heritable, of course, it must happen in a cell that will give rise to a gamete.)

The movement of a transposable element can have a variety of consequences. For instance, a transposable element that “jumps” into a protein-coding sequence will prevent the production of a normal transcript of the gene. (Introns provide a sort of “safety zone” that does not affect the transcript because the transposable element will be spliced out—unless it affects the splicing process.) If a transposable element inserts within a regulatory sequence, the transposition may lead to increased or decreased production of one or more proteins. Transposition caused both types of effects on the genes coding for pigment-synthesizing enzymes in McClintock’s corn kernels. Again, while such changes are usually harmful, in the long run some may provide a survival advantage. A possible example was mentioned earlier: At least some of the *Alu* transposable elements in the human genome are known to produce RNAs that regulate expression of human genes.

During transposition, a transposable element may carry along a gene or even a group of genes to a new position in the genome. This occurrence probably accounts for the location of the α-globin and β-globin gene families on different human chromosomes, as well as the dispersion of the genes of certain other gene families. By a similar tag-along process, an exon from one gene may be inserted into another gene in a mechanism similar to that of exon shuffling during recombination. For example, an exon may be inserted by transposition into the intron of a protein-coding gene. If the inserted exon is retained in the RNA transcript during RNA splicing, the protein that is synthesized will have an additional domain, which may confer a new function on the protein.

Most often, the processes discussed in this section produce harmful effects, which may be lethal, or have no effect at all. In a few cases, however, small heritable changes that are beneficial may occur. Over many generations, the resulting genetic diversity provides valuable raw material for natural selection. Diversification of genes and their products is an important factor in the evolution of new species. Thus, the accumulation of changes in the genome of each species provides a record of its evolutionary history. To read this record, we must be able to identify genomic changes. Comparing the genomes of different species allows us to do that, increasing our understanding of how genomes evolve. You will learn more about these topics next.